

ATPL EASA STUDY GUIDE | 031 - MASS AND BALANCE

Chapter 4: Mass Shifts, Load Changes & Ballast Calculations

Syllabus Source: CAE Oxford (pp. 67-88) | Pass Target: $\geq 90\%$ AviationExam

1. THE FUNDAMENTAL MASS SHIFT FORMULA

When an item of mass already on board is moved from one compartment to another, the aircraft's Total Mass DOES NOT CHANGE, but the Center of Gravity shifts in the direction of the movement!

- The Universal Mass Shift Formula: $(\text{Mass Shifted (m)}) / (\text{Total Aircraft Mass (M)}) = (\text{Change in CG (Delta CG)}) / (\text{Distance Shifted (d)})$
- Solving for Delta CG: $\text{Delta CG} = (m \times d) / M$
- Solving for Mass to Shift (m): $m = (M \times \text{Delta CG}) / d$
- Solving for Distance to Shift (d): $d = (M \times \text{Delta CG}) / m$

2. ADDING OR REMOVING MASS

When mass is added or removed, Total Mass CHANGES. Use the Moment equation:

- New CG Equation: $\text{New CG} = (\text{Old Total Moment} \pm \text{Added/Removed Moment}) / (\text{Old Total Mass} \pm \text{Added/Removed Mass})$

3. BALLAST CALCULATIONS (CORRECTING OUT-OF-LIMITS CG)

If an empty or lightly loaded ferry flight has a CG aft of the aft limit, ballast mass must be loaded into a forward cargo bay to bring the CG exactly onto the limit:

- Ballast Formula: $\text{Ballast Mass (m}_b) = (\text{Total Mass} \times (\text{Current CG} - \text{Desired Limit CG})) / (\text{Desired Limit CG} - \text{Ballast Arm})$

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Mass Shift & Ballast Formula

Problem 1 (Mass Shift): Aircraft mass 60,000 kg with CG at 15.00 m. Shift 1,200 kg from forward hold (Arm 10 m) to aft hold (Arm 30 m). - Step 1: Distance moved $d = 30 - 10 = 20$ meters. - Step 2: $\text{delta CG} = (\text{mass shifted} \times d) / \text{Total Mass} = (1,200 \times 20) / 60,000 = 0.40$ meters. - Step 3: Cargo moved aft $\rightarrow$ CG moves aft $\rightarrow$ New CG = $15.00 + 0.40 = 15.40$ meters. Problem 2 (Ballast Required): $\text{Mass Ballast} = [\text{Initial Mass} \times (\text{Target CG} - \text{Current CG})] / (\text{Ballast Arm} - \text{Target CG})$.

4. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Cargo Shift to Bring CG to Forward Limit

SCENARIO (Extremely Common AviationExam Question): An aircraft has: - Total Mass (M): 60,000 kg - Current CG position: 22.0 meters aft of datum - Certified Aft CG Limit: 21.6 meters aft of datum (The aircraft is 0.4 m OUT OF LIMITS!) - Forward Cargo Hold is at: 10.0 meters aft of datum - Aft Cargo Hold is at: 30.0 meters aft of datum QUESTION: How much cargo mass must be shifted from the Aft Cargo Hold to the Forward Cargo Hold to bring the CG to the Aft Limit of 21.6 m? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Identify all known variables: - Total Mass (M) = 60,000 kg - Desired change in CG (Delta CG) = Current CG - Target CG = 22.0 m - 21.6 m = 0.40 meters. - Distance of shift (d) = Distance from Aft Hold to Fwd Hold = 30.0 m - 10.0 m = 20.0 meters. Step 2: Recall the Master Shift Formula: - $m / M = \Delta \text{CG} / d$ -> $m = (M \times \Delta \text{CG}) / d$ Step 3: Plug in the numbers and calculate: - $m = (60,000 \text{ kg} \times 0.40 \text{ m}) / 20.0 \text{ m}$ - $m = 24,000 / 20.0 = 1,200 \text{ kg}$. FINAL ANSWER: You must shift exactly 1,200 kg of cargo from the Aft Hold to the Forward Hold!

[!] EXAM TRAP (AVIATIONEXAM) - The Direction of Shift Trap

- Shifting mass FORWARD moves the aircraft CG FORWARD (decreases CG distance from nose). - Shifting mass AFT moves the aircraft CG REARWARD (increases CG distance from nose). - Distance of shift (d) is ALWAYS the positive physical distance between the two compartments: $d = |\text{Hold}_A - \text{Hold}_B|$.

5. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: An aircraft has a gross mass of 5,000 kg and CG at station 120 inches. If 100 kg of baggage is shifted from

- [A]: 118 inches
- [B]: 115 inches
- [C]: 122 inches
- [D]: 110 inches

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [A]. Shift distance $d = 150 - 50 = 100$ inches (forward). $\Delta \text{CG} = (m \times d) / M = (100 \text{ kg} \times 100 \text{ in}) / 5,000 \text{ kg} = 10,000 / 5,000 = 2.0$ inches forward. New CG = 120 - 2.0 = 118 inches.

Q2: In mass shift calculations, what happens to the total mass of the aircraft?

- [A]: It increases by the amount of mass moved
- [B]: It decreases
- [C]: It remains EXACTLY CONSTANT
- [D]: It shifts by Delta CG

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. Moving an object from one cabin location to another redistributes internal forces but does not add or subtract mass. Total aircraft mass M remains constant.

Q3: If 500 kg of cargo is REMOVED from a cargo bay located forward of the aircraft's current CG, what will happen to the CG?

- [A]: It will move forward
- [B]: It will move REARWARD (aft)

- [C]: It will remain stationary
- [D]: It becomes negative

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Removing mass from in front of the CG takes away forward weight, causing the remaining balance to tilt backwards (CG moves aft).